

Lepton Residuals: Lepton Ladder, Galois and Koide Compared

How Empirical Values Arise, Where Theory and Irrational Factors Enter, and What
This Means for FFGFT Residuals

Johann Pascher
ORCID 0009-0000-6518-4064

5 September 2026

Contents

Lepton Residuals: Lepton Ladder, Galois and Koide Compared	1
1 Mass Formula and Quantum Numbers	1
2 Lepton Ladder Predictions	2
3 Galois Identities	2
4 Koide Relation	2
5 Overall Comparison	3
6 Findings	3
7 Scale Invariance and Its Consequences	3
8 Koide Condition in the FFGFT Quantum Numbers	4
9 Generation-Dependent Corrections and Koide	6
10 Why Do ε_i Decrease with Generation?	7

Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- QED self-energy: $\frac{3\alpha}{4\pi} \ln(m_i/\mu) \rightarrow \frac{3\alpha}{4\pi} \ln(m_i^2/\mu^2)$.
- Koide angle (1 Oct. 2026): "all three agree to $\leq 0.5 \cdot \xi'' \rightarrow$ PDG and A110 agree to $< 0.1 \cdot \xi$, the FFGFT value lies $-9 \cdot \xi$ off (section on the angle θ and conclusion).

Status Markers

- [K]** *Confirmed* — derived from ξ , numerically verified.
- [B]** *Proved* — algebraically established.
- [S]** *Speculative* — open.
- [Q]** *Source* — corpus-external primary source, verified here.

Abstract

Three independent FFGFT structures make predictions for lepton ratios: the lepton ladder (Docs. 006/317), the Galois identities (Doc. 351), and the Koide relation. This document compares all three predictions with PDG values and expresses residuals in units of $\xi = 4/30000$. All formulae are exactly rational; numerical evaluations use $m_e = 0.51099895$ MeV, $m_\mu = 105.6583755$ MeV, $m_\tau = 1776.86$ MeV (PDG) **[Q]**.

Theoretical Claim

FFGFT claims predictions for dimensionless ratios, not for dimensional individual masses (Doc. 351, § Theoretical Claim). Individual masses carry K_{frak} , SI projection, and QED-dependent extraction; these produce an unavoidable bandwidth of order ξ to 100ξ . Ratios, by contrast, are correction-free — *provided the corrections are uniform* (see § 9).

1 Mass Formula and Quantum Numbers

The FFGFT mass formula reads (Docs. 006, 333):

$$m_i = r_i \cdot \xi^{p_i} \cdot \nu \cdot K_{\text{frak}} \quad [\mathbf{K}] \quad (1)$$

with $\xi = 4/30000$ **[B]**, $K_{\text{frak}} = 74/75$ **[K]**. In the ratio of two leptons ν and K_{frak} cancel:

$$\frac{m_j}{m_i} = \frac{r_j}{r_i} \cdot \xi^{p_j - p_i} \quad [\mathbf{B}] \quad (2)$$

Particle	r_i	p_i	n_θ, n_ϕ	$r_i = n_\phi^2/n_\theta$
Electron e	4/3	3/2	3, 2	4/3
Muon μ	16/5	1	5, 4	16/5
Tau τ	25/9	2/3	9, 5	25/9

2 Lepton Ladder Predictions

$$\frac{m_\mu}{m_e} = \frac{16/5}{4/3} \cdot \xi^{1-3/2} = \frac{12}{5} \cdot \xi^{-1/2} = \frac{12}{5} \cdot \sqrt{7500} \quad (3)$$

Ratio	Exact formula	FFGFT	PDG	Residual
m_μ/m_e	$\frac{12}{5} \cdot \xi^{-1/2}$	207.846	206.768	$+39.1 \cdot \xi$ [K]
m_τ/m_μ	$\frac{125}{144} \cdot \xi^{-1/3}$	16.992	16.817	$+77.9 \cdot \xi$ [K]
m_τ/m_e	$\frac{25}{12} \cdot \xi^{-5/6}$	3531.6	3477.2	$+117.4 \cdot \xi$ [K]

Algebraic dependence **[B]**:

$$\varepsilon(\tau/e) \approx \varepsilon(\tau/\mu) + \varepsilon(\mu/e) = 77.9 + 39.1 = 117.0 \cdot \xi \quad (4)$$

(The exact value 117.4 contains a cross-term of order ξ^2 .) There are only two independent residuals.

3 Galois Identities

From the Galois field structure (Doc. 351) **[B]**:

$$(m_\mu/m_e)^2 = 43200 = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| \quad (5)$$

$$m_e \cdot m_\mu = 54 \text{ MeV}^2 \quad (\text{Layer-2 with anchor } m_e) \quad (6)$$

Identity	Galois	PDG	Residual	Status
$(m_\mu/m_e)^2$	43200	42753.1	$-77.6 \cdot \xi$	[K]; cause [S]
$m_e \cdot m_\mu$ [MeV ²]	54	53.991	$-1.21 \cdot \xi$	emp. discrepancy [K]

Connection to the lepton ladder. The Galois difference residual $-77.6 \cdot \xi$ equals twice the linear lepton ladder residual ($+39.1 \cdot \xi$) with opposite sign — algebraically forced, since $(m_\mu/m_e)_{\text{FFGFT}}^2 > (m_\mu/m_e)_{\text{PDG}}^2$ if and only if $m_\mu/m_e|_{\text{FFGFT}} > m_\mu/m_e|_{\text{PDG}}$.

4 Koide Relation

$$Q \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} \stackrel{?}{=} \frac{2}{3} \quad (7)$$

Input	Q	Deviation	Status
PDG masses	0.666 660 5	$-0.046 \cdot \xi$	[K]
FFGFT predictions (anchor m_e)	0.667 742	$+8.06 \cdot \xi$	numerical

Koide as m_τ prediction. Given PDG m_e , m_μ and $Q = 2/3$, algebraically **[K]**:

$$m_{\tau, \text{Koide}} = 1776.969 \text{ MeV} \quad \text{vs.} \quad m_{\tau, \text{PDG}} = 1776.860 \text{ MeV} \quad \text{deviation: } +0.46 \cdot \xi \quad (8)$$

This is the most precise prediction in the entire comparison.

5 Overall Comparison

Quantity	Method	Prediction	PDG	Residual	Status
m_μ/m_e	Ladder	207.846	206.768	$+39.1 \cdot \xi$	[K]
m_τ/m_μ	Ladder	16.992	16.817	$+77.9 \cdot \xi$	[K]
m_τ/m_e	Ladder	3531.6	3477.2	$+117.4 \cdot \xi$	[K]; sum
$(m_\mu/m_e)^2$	Galois	43200	42753	$-77.6 \cdot \xi$	[K]; cause [S]
$m_e \cdot m_\mu$ [MeV ²]	Galois	54	53.991	$-1.21 \cdot \xi$	emp. disc. [K]
Q_{Koide}	Koide	2/3	0.66666	$-0.046 \cdot \xi$	[K]
m_τ [MeV]	Koide	1776.97	1776.86	$+0.46 \cdot \xi$	[K]

6 Findings

(1) Koide is most precise [K]. PDG masses satisfy Koide to $0.046 \cdot \xi$. Koide gives with $+0.46 \cdot \xi$ the most accurate m_τ prediction in the comparison. FFGFT predictions meet Koide only to $+8 \cdot \xi$.

(2) Lepton ladder and Galois share the same basic residual [K]. The Galois difference residual ($-77.6 \cdot \xi$) equals exactly $-2 \times (+39.1 \cdot \xi)$ of the lepton ladder residual for m_μ/m_e . This is algebraically forced.

(3) All residuals positive except Galois [K]. Lepton ladder and Koide- m_τ lie just above PDG; Galois- $(m_\mu/m_e)^2$ and Q_{Koide} lie just below target. Signs are consistent.

(4) Range of residuals [K].

Method	Residual
Koide (Q)	$\approx 0.05 \cdot \xi$
Galois (product)	$\approx 1 \cdot \xi$
Koide (m_τ)	$\approx 0.5 \cdot \xi$
Lepton ladder (m_μ/m_e) / Galois (square)	$\approx 39-78 \cdot \xi$

(5) Koide is scale-invariant [B]. For every $\lambda > 0$:

$$Q(\lambda m_1, \lambda m_2, \lambda m_3) = Q(m_1, m_2, m_3) \quad \text{[B]} \quad (9)$$

Consequence: no uniform factor — neither v , nor K_{frak} , nor a common SI projection — can change Q . Option 2 (corrections improve Koide) is thus algebraically excluded.

7 Scale Invariance and Its Consequences

What Koide can measure

Q depends exclusively on the ratios of masses, not on their absolute value:

$$Q = Q\left(1, \frac{m_\mu}{m_e}, \frac{m_\tau}{m_e}\right) \quad (10)$$

With the FFGFT ratios $(m_\mu/m_e)_{\text{FFGFT}} = 207.85$ and $(m_\tau/m_e)_{\text{FFGFT}} = 3531.6$, $Q_{\text{FFGFT}} = 0.66774$ is fully determined — independent of v , K_{frak} , or the SI projection.

Three tested options

Option	Q change	Status
Uniform K_{frak} (all equal)	none	[B] excluded
Single v anchor	none	[B] excluded
Generation-dep. $K_{\text{frak}}^{\alpha p_i}$	yes, but worse masses	fit without physics

Generation-dependent test. The exponent α in $m_i \cdot K_{\text{frak}}^{\alpha p_i}$ that forces $Q = 2/3$ lies at $\alpha \approx -36/17$. The resulting masses ($m_e = 0.527$ MeV, $m_\mu = 108.0$ MeV, $m_\tau = 1817.6$ MeV) deviate further from PDG than the uncorrected bare masses. No physics behind this value.

Conclusion

Koide is not contained in the FFGFT lepton ladder **[S]**. The lepton ladder organises masses via logarithmic ξ powers: $m_i \propto \xi^{p_i}$. Koide organises masses via square roots $\sqrt{m_i}$. These two structures are orthogonal — there is no algebraic relation between ξ^{p_i} and $\sqrt{r_i \xi^{p_i}}$ that forces Koide.

PDG masses satisfy Koide to $0.046 \cdot \xi$ because the three PDG masses as a measurement system are mutually consistent — they carry implicitly a symmetry not yet anchored in the FFGFT quantum numbers (r_i, p_i) .

The natural candidate for this anchoring is the Z_3 circulant structure of the lepton mass operator (Doc. A110): there $\theta = 2/9$ describes the Koide phase in the angular sector.

8 Koide Condition in the FFGFT Quantum Numbers

Z_3 parametrisation

The eigenvalues of a Z_3 circulant mass operator can be written as:

$$\sqrt{m_k} = c + d \cdot \cos\left(\theta + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2 \quad (11)$$

Then $\sum \sqrt{m_k} = 3c$ and $\sum m_k = 3c^2 + \frac{3}{2}d^2$, giving:

$$Q = \frac{1}{3} + \frac{d^2}{6c^2} \quad (12)$$

$$Q = \frac{2}{3} \iff d = c\sqrt{2} \quad \textbf{[B]} \quad (13)$$

d/c ratio in PDG and FFGFT

Source	d/c	$d/c - \sqrt{2}$	Status
PDG	1.41420	$-0.098 \cdot \xi$	[K]
FFGFT predictions	1.41649	$+17.09 \cdot \xi$	numerical
Target	$\sqrt{2} = 1.41421$	0	—

PDG satisfies $d = c\sqrt{2}$ to $0.10 \cdot \xi$. FFGFT predictions miss it by $+17 \cdot \xi$.

Angle θ and connection to Doc. A110

With assignment $k = 0 \rightarrow \tau$, $k = 1 \rightarrow e$, $k = 2 \rightarrow \mu$:

Source	θ [rad]	θ [°]
PDG	0.22223	12.733
FFGFT predictions	0.22099	12.662
Doc. A110 ($\theta = 2/9$)	0.22222	12.732

PDG and A110 agree to $< 0.1 \cdot \xi$; the FFGFT value lies $-9 \cdot \xi$ off. (*Corrected on 1 Oct. 2026: previously "All three agree to $\lesssim 0.5 \cdot \xi$ [K]" — $(0.22223 - 2/9)/\xi = +0.06$, but $(0.22099 - 2/9)/\xi = -9.2$.)* The Koide phase angle $\theta = 2/9$ from Doc. A110 correctly describes the angular structure. The amplitude d/c , however, is not determined by θ — it is an independent condition.

Koide condition in the quantum numbers (r_i, p_i)

Since $x_i = \sqrt{r_i} \cdot \xi^{p_i/2}$ and Koide is scale-invariant, Q depends only on the differences of exponents:

$$\Delta p_e = p_e - p_\tau = \frac{5}{6}, \quad \Delta p_\mu = p_\mu - p_\tau = \frac{1}{3}, \quad \Delta p_\tau = 0 \quad (14)$$

The exact Koide condition $Q = 2/3$ requires:

$$r_{\mu, \text{Koide}} = 3.2378 \dots \quad \text{instead of} \quad r_\mu = \frac{16}{5} = 3.2000 \quad (15)$$

Deviation: $\Delta r_\mu / r_\mu = +88.6 \cdot \xi$. The nearest rational approximation is $531/164$ — no representation as n_ϕ^2/n_θ with small integer winding numbers.

Particle	FFGFT r_i	Winding (n_θ, n_ϕ)	Koide- r_i
Electron	4/3	(3, 2)	4/3 (fixed)
Muon	16/5	(5, 4)	3.2378...
Tau	25/9	(9, 5)	25/9 (fixed)

The Koide- r_μ has no clean torus winding structure **[S]**.

Orthogonality of θ and d/c **[B]**

The Z_3 circulant has exactly two independent real parameters: the angle θ (rotation in the Z_3 space) and the ratio d/c (eccentricity). Their orthogonality is algebraically proved: $Q = \frac{1}{3} + \frac{d^2}{6c^2}$ does *not* contain θ — the Koide condition is exclusively a statement about d/c , independent of θ **[B]**.

Consequence for FFGFT: the corpus delivers $\theta = 2/9$ from the geometry of the Z_3 circulant (Doc. A110) **[K]**. The condition $d/c = \sqrt{2}$ is an *additional, independent* requirement — the R113 bridge. Both together would be needed to derive Koide fully from FFGFT. Currently only θ is anchored; $d/c = \sqrt{2}$ remains open **[S]**.

Conclusions of the test

1. **Angle $\theta = 2/9$ **[K]**:** The angular structure of the Z_3 circulant agrees between PDG and Doc. A110 to $< 0.1 \cdot \xi$; the FFGFT value lies $-9 \cdot \xi$ off. (*Corrected on 1 Oct. 2026: previously "between PDG, FFGFT and Doc. A110 to $\lesssim 0.5 \cdot \xi$ " — see the section on the angle θ .*)
2. **Amplitude d/c — two distinct objects: **[K]**:** given the observed generation-dependent ε_i , their effect on the Koide amplitude is calculable and brings the bare FFGFT d/c ($+17 \cdot \xi$ off $\sqrt{2}$) quantitatively to $\sqrt{2}$ — 101 % explained (see § 9). **[S]**: the FFGFT-native derivation of *why* the ε_i possess this generation-dependent structure — and therefore why $d/c = \sqrt{2}$ should arise from the T^4/Z_3 geometry — remains open (R113 bridge, § 10).

9 Generation-Dependent Corrections and Koide

The ε_i are not uniform

The bare FFGFT masses deviate from PDG masses by generation-dependent residuals ε_i :

$$m_i^{\text{PDG}} = m_i^\circ \cdot (1 + \varepsilon_i) \quad \Longleftrightarrow \quad \varepsilon_i = \frac{m_i^{\text{PDG}}}{m_i^\circ} - 1 \quad \mathbf{[K]}$$

The ε_i are *computed residuals* from PDG **[Q]** against FFGFT bare **[K]** — not numbers taken from the QED literature. The column "Presumed origin" is an attribution **[S]**, not a citation.

Particle	ε_i	Presumed origin [S]
Electron	$+192 \cdot \xi$	SI projection, QED
Muon	$+152 \cdot \xi$	SI projection, QED extraction (R113)
Tau	$+73 \cdot \xi$	SI projection, measurement scatter

The ε_i decrease with generation — they are not uniform.

Why non-uniform ε_i shift Koide

Koide is scale-invariant under uniform λ **[B]**. For different ε_i , to first order:

$$\Delta Q \approx Q^\circ \cdot (\langle \varepsilon \rangle_m - \langle \varepsilon \rangle_{\sqrt{m}}) \quad (16)$$

Since $\varepsilon_\tau < \varepsilon_\mu < \varepsilon_e$ and m_τ dominates the mass sum, $\langle \varepsilon \rangle_m < \langle \varepsilon \rangle_{\sqrt{m}}$, hence $\Delta Q < 0$.

Numerical result

Quantity	Value	Status
$Q(\text{bare}) - 2/3$	$+8.06 \cdot \xi$	from quantum numbers
$\langle \varepsilon \rangle_m$	$+77.7 \cdot \xi$	mass-weighted
$\langle \varepsilon \rangle_{\sqrt{m}}$	$+90.0 \cdot \xi$	$\sqrt{m}$ -weighted
ΔQ	$-8.23 \cdot \xi$	perturbation theory
$Q(\text{bare}) + \Delta Q - 2/3$	$-0.17 \cdot \xi$	estimate
$Q(\text{PDG}) - 2/3$	$-0.05 \cdot \xi$	exactly measured
Explanatory fraction	101 %	[K]

Revised conclusion **[K]**

The statement "ratios are correction-free" holds for uniform corrections **[B]**. For Koide additionally:

- Simple ratios m_j/m_i are correction-free under uniform λ **[B]**.
- Koide Q responds to non-uniform ε_i because it is defined via $\sqrt{m_i}$.
- The generation-dependent ε_i (order 73–192 $\cdot \xi$, decreasing with generation) explain the transition $Q_{\text{bare}}(+8 \cdot \xi) \rightarrow Q_{\text{PDG}}(-0.05 \cdot \xi)$ to 101 % **[K]**.
- The remaining open point is: *why do ε_i decrease with generation?* The answer lies in the QED extraction (R113) and SI projection (Docs. 085, 333) — their generation-dependent structure is analysed in § 10.

10 Why Do ε_i Decrease with Generation?

*Status separation in this section developed in correspondence with Stefaan Vossen, applying the Constitutional Physics / Canonical Constitutional Record framework (Doc. 351, 5–6 September 2026): the calculable effect of the observed ε_i **[K]** is distinguished from their source-native derivation **[S]**.*

Two sources of non-uniform corrections

1. QED self-energy (R113) [Q]. The measured pole mass of a charged lepton differs from the bare mass by the QED self-energy. At leading order:

$$\frac{\delta m_i}{m_i} \approx \frac{3\alpha}{4\pi} \ln\left(\frac{m_i^2}{\mu^2}\right) + \text{const} \quad (17)$$

(Corrected on 30 Sept. 2026: previously $\frac{3\alpha}{4\pi} \ln(m_i/\mu)$ – the one-loop self-energy coefficient $3\alpha/4\pi$ belongs to $\ln(m^2/\mu^2)$; with $\ln(m/\mu)$ it reads $3\alpha/2\pi$.) where μ is the renormalisation scale. Since $m_\tau \gg m_\mu \gg m_e$, $\ln(m_i/\mu)$ grows with generation.

The observed trend $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$ is, however, *opposite* to the naive QED self-energy that grows with mass. The reason: ε_i measures the gap bare FFGFT mass $\rightarrow$ PDG mass. This gap is a *difference* of two quantities each scaling with mass. The SI projection contributes more strongly at small masses (relative correction $\propto 1/m_i$ for a fixed absolute projection uncertainty).

2. SI projection (Docs. 085, 333) [S]. The SI projection maps the bare FFGFT mass $m_i^\circ = r_i \cdot \xi^{p_i} \cdot v \cdot K_{\text{frak}}$ to the measured MeV value. The factor $K_{\text{frak}} = 74/75$ is uniform **[B]** and does not shift Q (§ 5). The *non-uniform* component arises from the mass-dependent QED extraction: the experimental determination of m_μ from muon decay and of m_τ from τ decay widths each involve different orders of perturbation theory (R113). These are of order $\alpha \approx 1/137$, hence of order ξ to 100ξ — consistent with the observed ε_i .

What FFGFT can derive and what it cannot

Quantity	Status	Origin
Bare masses $m_i^\circ = r_i \cdot \xi^{p_i} \cdot v \cdot K_{\text{frak}}$	[K]	FFGFT internal
Uniform factor K_{frak}	[B]	FFGFT internal
Order of magnitude of ε_i ($\sim \alpha/\pi$, order ξ – 100ξ)	[K]	QED external [Q]
Falling structure $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$	[S]	QED + SI projection, not derived from FFGFT quantum numbers
QED self-energy from T^4/Z_3 geometry	[S]	open bridge R113

Connection to Doc. 351

The same non-uniform ε_i structure explains the ratio deviations in Doc. 351: m_μ/m_e deviates by $\sim 39 \cdot \xi$ and m_τ/m_μ by $\sim 78 \cdot \xi$ from FFGFT predictions. Simple ratios are correction-free under *uniform* λ **[B]**; under non-uniform ε_i they carry the difference residual $\varepsilon_j - \varepsilon_i$. The bridge R113 (QED extraction, mass-dependent) is thus the common open point of Docs. 351 and 352 (Doc. 351, § 5: "R113: m_μ/m_e QED-dependent", text correction "model-independent" $\rightarrow$ "least model-dependent").

Conclusion

The observed falling ε_i structure is consistent with a combined QED/SI-projection account, but its derivation from FFGFT quantum numbers remains open **[S]**. FFGFT delivers the bare masses; the spread of ε_i is an external **[Q]** fact in its order of magnitude. Whether the QED self-energy is derivable from the T^4/Z_3 geometry remains open **[S]** — this is the R113 bridge.

Verification script: `pruef_352_leptonreste.py` — all numbers reproducible, pure Python standard library.

Sources

PDG 2024 **[Q]**. Doc. 006 (mass formula), Doc. 317 (lepton ladder values), Doc. 351 (Galois identities, ε_i values, R113), Doc. A110 (Z_3 circulant), Docs. 085, 333 (SI projection), R113 (QED dependence m_μ/m_e), Doc. 190/R114 (register entry for this document).